An Analysis of Residual connection neural network

Residual connection neural network

$$a^{(l)} = \sigma! \, (W^{(l)} a^{(l-1)} + b^{(l)})$$

Residual connection neural network -- Part 1

where φ {\displaystyle \phi } can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

Residual connection neural network -- Part 2

A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of x ■ f (x) + x {\displaystyle x\mapsto f(x)+x} , where f {\displaystyle f} is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

Residual connection neural network -- Part 3

x ■ + 2 = F (x ■ + 1) + x ■ + 1 = F (x ■ + 1) + F (x ■) + x ■ {\displaystyle {\begin{aligned}x_{\ell +2}&=F(x_{\ell +1})+x_{\ell +1}\\&=F(x_{\ell +1})+F(x_{\ell })+x_{\ell }\end{aligned}}}

Residual connection neural network -- Part 4

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c t {\displaystyle c_{t}} that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

Residual connection neural network -- Part 5

∂ E ∂ x ■ = ∂ E ∂ x L ∂ x L ∂ x ■ = ∂ E ∂ x L (1 + ∂ ∂ x ■ ∑ i = ■ L − 1 F (x i)) = ∂ E ∂ x L + ∂ E ∂ x L ∂ ∂ x ■ ∑ i = ■ L − 1 F (x i) {\displaystyle {\begin{aligned}{\frac {\partial {\mathcal {E}}}{\partial x_{\ell }}}&={\frac {\partial {\mathcal {E}}}{\partial x_{L}}}{\frac {\partial x_{L}}{\partial x_{\ell }}}\\&={\frac {\partial {\mathcal {E}}}{\partial x_{L}}}\left(1+{\frac {\partial }{\partial x_{\ell }}}\sum _{i=\ell }^{L-1}F(x_{i})\right)\\&={\frac {\partial {\mathcal {E}}}{\partial x_{L}}}+{\frac {\partial {\mathcal {E}}}{\partial x_{L}}}{\frac {\partial }{\partial x_{\ell }}}\sum _{i=\ell }^{L-1}F(x_{i})\end{aligned}}}

Residual connection neural network -- Part 6

Residual connection In a multilayer neural network model, consider a (non-residual) subnetwork with a certain number of stacked layers (e.g., 2 or 3). Let H (x ; α) {\displaystyle H(x;\alpha)} denote the subnetwork. Suppose H ∗ {\displaystyle H^{*}} is the desired optimal output of this subnetwork. Residual learning simply adds x {\displaystyle x} directly to the output, such that the optimal learned output now becomes be H ∗ − x {\displaystyle H^{*}-x} , which is interpreted as a "residual" with respect to x {\displaystyle x} . The operation of "adding x {\displaystyle x} " is implemented via a "skip connection" that performs an identity mapping to connect the input of the subnetwork with its output. This connection is referred to as a "residual connection" in later work. Let F (x ; α) = H (x ; a) + x {\displaystyle F(x;\alpha)=H(x;a)+x} . The function F {\displaystyle F} is often represented by matrix multiplication interlaced with activation functions and normalization operations (e.g., batch normalization or layer normalization). As a whole, one of these subnetworks is referred to as a "residual block". A deep residual network is constructed by simply stacking these blocks. Long short-term memory (LSTM) has a memory mechanism that serves as a residual connection. In an LSTM without a forget gate, an input x t {\displaystyle x_{t}} is processed by a function F {\displaystyle F} and added to a memory cell c t {\displaystyle c_{t}} , resulting in c t + 1 = c t + F (x t) {\displaystyle c_{t+1}=c_{t}+F(x_{t})} . An LSTM with a forget gate essentially functions as a highway network. To stabilize the variance of the layers' inputs, it is recommended to replace the residual connections x + f (x) {\displaystyle x+f(x)} with x / L + f (x) {\displaystyle x/L+f(x)} , where L {\displaystyle L} is the total number of residual layers.

Residual connection neural network -- Part 7

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form x ■ F (x) + P (x) {\displaystyle x\mapsto F(x)+P(x)} , where P {\displaystyle P} is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".

Residual connection neural network -- Part 8

During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.